

Magnetic Hard Disk Mechanism

Basic Structure of a Hard Disk

- **Platters:** Circular disks coated with magnetic material, used to store data.
- **Read-Write Head:** Electromagnetic device that reads/writes data onto platter surfaces.
- **Disk Arm:** Moves the head radially across platters to access different tracks.
- **Spindle:** Rotates all platters at high speed (e.g., 5400–15000 RPM).

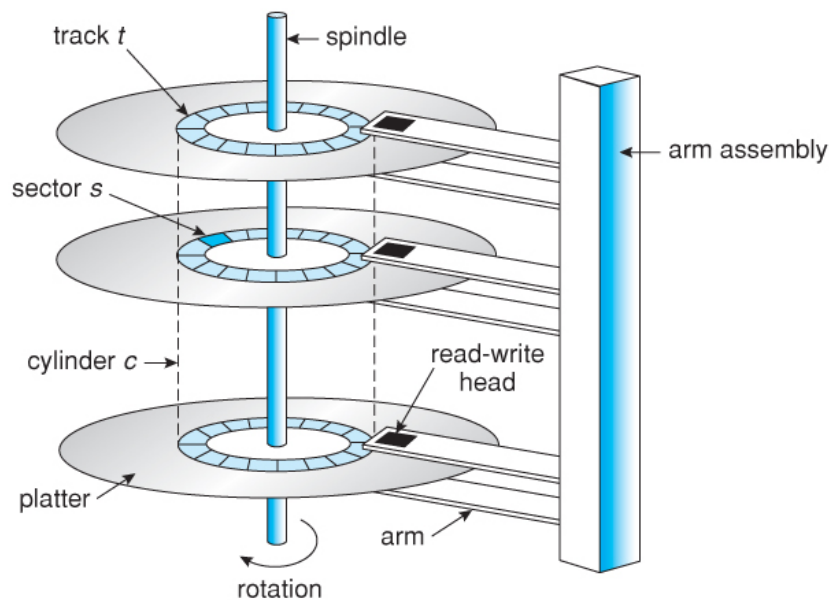


Figure 1: Schematic of a magnetic hard disk with platters, disk arm, and read-write heads.

Data Organization on Platters

- **Tracks:** Concentric circles on the platter surface. Typical hard disks contain **50K–100K tracks per platter**.
- **Sectors:** Each track is divided into **sectors**, the smallest readable/writable unit. Standard sector size = **512 bytes**.
- **Sectors per Track:**
 - Inner tracks: **500–1000 sectors**.
 - Outer tracks: **1000–2000 sectors**.
- **Cylinder:** The set of tracks with the same radius across all platters.

Read/Write Mechanism

1. Disk arm swings to position the head over the correct track (**seek operation**).
2. The platter spins continuously at high speed.
3. Data is read or written as the desired sector passes under the read-write head.

Head-Disk Assemblies

- Modern disks contain **multiple platters** mounted on a single spindle (usually 1–5).
- Each platter has a read-write head.
- All heads are mounted on a common arm, moving together across platters.

Disk Controller Functions

- Interfaces between the computer system and disk hardware.
- Accepts high-level commands (e.g., read/write sector).
- Initiates mechanical actions (moving arm, spinning platters).
- Computes and attaches **checksums** to each sector.
- Verifies data integrity during reads (mismatched checksum → corrupted data).
- Ensures successful writes by reading back the sector immediately.
- Handles **remapping of bad sectors** to maintain reliability.

Summary

- Hard disks store data magnetically on spinning platters.
- Data is organized into **tracks**, **sectors**, and **cylinders**.
- Access involves mechanical movement (**seek time**) and platter rotation (**rotational latency**).
- Disk controllers enhance performance and reliability through error-checking and bad-sector management.